

Computer Science Portfolio

Project Report

Maide Oskay

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Links

- **Portfolio website:** <https://maidesource.github.io>
- **GitHub repository:** <https://github.com/maidesource/maidesource.github.io>

1 Introduction

This report accompanies the computer science portfolio I built for a university course assignment, which simulates preparing a portfolio for a working student position application. The portfolio consists of a personal website hosted with GitHub Pages and its corresponding public GitHub repository, which together present my education, technical skills, and ongoing projects, alongside a LaTeX-typeset curriculum vitae. This document explains how the portfolio is structured, the reasoning behind its design, the tools used to build it, and a reflection on its current strengths, limitations, and planned improvements.

2 Repository and Website Structure

The repository is named `maidesource.github.io`, which GitHub automatically recognises as a user site and serves directly from the `main` branch at <https://maidesource.github.io>. The repository root is kept flat and predictable:

- `index.html`: the single-page website, containing markup for the header, About, Education, Projects, and Skills sections, and the footer.
- `styles.css`: all visual styling, including the color palette, typography, spacing, and the card-based layout used for project entries.
- `script.js`: a small script that enables smooth scrolling when the navigation links are clicked.
- `assets/`: the profile photograph used in the page header.
- `docs/`: the compiled CV (`cv.pdf`) together with its LaTeX source (`cv.tex`), so the CV can be regenerated or edited without leaving the repository.
- `projects/`: documentation of the technical exercise showcased in the portfolio. At this stage, the website itself is treated as the featured project, and this folder explains why and how it was built (see `projects/README.md`).
- `README.md`: a top-level description of the repository's purpose and layout for anyone visiting the project on GitHub.

On the website itself, the navigation bar links to in-page sections (About, Projects, Skills) using anchor links, and to the CV, which opens the PDF stored in `docs/`. This keeps the whole

experience of browsing the site and reading the CV inside a single, coherent repository, rather than splitting content across external services.

3 Design Decisions

Several deliberate choices shaped the final result:

- **Single-page layout.** A hiring manager should be able to see education, skills, and projects within a few seconds of scrolling, without navigating through multiple pages. A single page with anchored sections keeps the site quick to scan while still feeling organised.
- **Consistent visual identity.** The color palette (warm terracotta and beige tones) was carried over from my personal CV design so that the website and CV are immediately recognisable as belonging to the same person and brand.
- **Card-based project entries.** Each project is presented in its own visually distinct card with a title, date range, and bullet points. This mirrors how the same content is organised in the CV, so a reader who has seen one recognises the same information in the other.
- **Minimal dependencies.** The site intentionally avoids CSS or JavaScript frameworks. For a page of this size and complexity, a framework would add download weight and hide the underlying HTML/CSS/JS skills that the portfolio is meant to demonstrate in the first place.
- **Version-controlled CV.** Rather than treating the CV as an external, opaque PDF, its LaTeX source lives in the same repository. This makes the CV auditable, diffable, and easy to update the same way the rest of the site is updated: edit, commit, push.

4 Tools and Technologies

- **HTML5 and CSS3** for the page structure and styling, using semantic elements (`header`, `main`, `section`, `footer`) and Flexbox for layout.
- **Vanilla JavaScript (ES6)** for the smooth-scroll navigation behaviour, with no external libraries.
- **Git and GitHub** for version control, and **GitHub Pages** for hosting and continuous deployment: every push to `main` automatically republishes the live site.
- **LaTeX** (compiled with `pdflatex` on TeX Live) for typesetting the CV, allowing precise control over layout and easy long-term maintenance as a plain-text source file.
- A Linux terminal (Fedora) with `nano` was used throughout for editing files and running Git commands, reinforcing familiarity with a command-line development workflow.

5 Reflection

5.1 Strengths

- The site is lightweight and loads quickly, since it has no external framework dependencies.
- Content, styling, and behaviour are cleanly separated into `index.html`/`styles.css`/`script.js`, which keeps the codebase easy to read and extend.
- The CV is reproducible and version-controlled rather than being a static, unexplained binary file.

- The visual identity is consistent across the website and CV, giving a professional and cohesive impression.
- The deployment pipeline is simple and reliable: GitHub Pages removes the need for any separate hosting setup.

5.2 Weaknesses

- The layout is not yet fully responsive; it has not been adapted with media queries for small screens, so the experience on mobile devices is not optimised.
- The `projects/` folder currently documents the portfolio site itself rather than a separate body of coding exercises, since I do not yet have additional finished technical projects to include.
- There is no automated testing, linting, or build process; the site is edited and deployed manually.
- No accessibility audit (colour contrast, screen-reader behaviour) has been performed yet.

5.3 Planned Improvements

- Add responsive, mobile-first CSS using media queries so the site adapts cleanly to smaller screens.
- Populate `projects/` with additional, independent technical exercises (small scripts, algorithms, or coursework projects) as they are completed.
- Introduce basic automated checks (e.g. an HTML/CSS validator or a Lighthouse accessibility and performance audit) as part of the development workflow.
- Improve SEO and shareability with meta tags such as an Open Graph description and page summary.
- Consider a short "notes" or "learning log" section documenting technical topics studied during my degree, to keep the portfolio active between larger projects.

6 Conclusion

The portfolio in its current form fulfils its core purpose: it presents my background, skills, and ongoing projects clearly, is fully version-controlled, and is publicly accessible through GitHub Pages. Its main limitations, namely a thin project section and a non-responsive layout, are well understood and have a concrete improvement plan, which I intend to carry out as I complete further coursework and personal projects.